

Azure Network Prototype

PrimeCore Minerals | Secure Multi Office Network

PC-05 | Version 0.1 | 23 September 2026

Status: Proposed prototype not deployed

Owner: Cloud Administrator | Approver: Project lead | Approval date: Pending

Provide a bounded Azure design for testing remote access to a private PrimeCore workload. The prototype is separate from the evidenced HQ lab and does not imply an existing cloud deployment.

Purpose and design decision

Use one virtual network, a route based VPN gateway and one private test VM. This demonstrates controlled remote access without publishing VM management ports to the Internet. Use disposable data. HQ or mine-site connectivity is outside this prototype until a separate site-to-site design is approved.

Resource	Proposed value	Purpose
Resource group	rg-pc-network-lab-aue	Dedicated lifecycle and cost boundary
Region	Australia East	Project design assumption; verify subscription availability
Virtual network	vnet-pc-lab 10.60.0.0/16	Separate from known HQ ranges
GatewaySubnet	10.60.255.0/27	Reserved gateway subnet; no NSG
Workload subnet	snet-workload 10.60.10.0/24	Private test VM and workload NSG
VPN client pool	172.30.250.0/24	Distinct remote-client addresses
Test VM	vm-pc-test-01; proposed private IP 10.60.10.10	No public IP; HTTPS test service
VPN gateway	vgw-pc-lab; route based; supported non-Basic SKU	OpenVPN with Microsoft Entra ID

Verify these ranges against all branch networks and remote clients before deployment. The /27 GatewaySubnet follows Microsoft guidance; do not attach an NSG to it. Select the currently available gateway SKU only after capability and budget review. [M1]

Security boundaries

On the workload NSG, allow P2S pool TCP 443 to the test VM. Permit the chosen management port only from an approved administrator path. Add an explicit deny for remaining P2S-to-workload traffic at a lower priority than the specific allows, before the default virtual-network allow. Apply the same intent at the guest firewall. A VPN sign-in alone does not grant application access.

Build and acceptance plan

1. Record subscription, tenant, authorised deployer, resource owner, expiry and budget. Set cost alerts and tags for Project=PrimeCore, Environment=Lab, Owner and Expiry. Alerts do not automatically stop spending.
2. Create the resource group and VNet with only the planned subnets. Verify overlap and route tables. Create the workload NSG and explicit allows and denies.
3. Create the route based VPN gateway and required public IP resource using the selected SKU. Leave BGP and site-to-site connections disabled for this prototype.
4. Configure P2S using PC-15. Use the tenant-specific authority, audience and issuer from current Microsoft guidance and the Microsoft-registered client. Require authorised users and test the intended MFA or Conditional Access policy. [M2]
5. Create the private VM and test service. Use secure credential provisioning. Configure diagnostic collection and a restricted destination, then record effective routes and NSG rules.
6. Test connectivity with an authorised client and denial with an unauthorised identity. Verify the application port, a prohibited port, disconnection, and the absence of a public VM management endpoint.

Acceptance test	Required evidence
AZ-01 resources match design	Resource export, addresses, tags and gateway settings
AZ-02 authorised VPN and HTTPS access	Assigned client IP, route, identity result and service response
AZ-03 unauthorised user or prohibited service	Denied sign-in or connection with matching evidence
AZ-04 logging and shutdown	Diagnostic retrieval and owner-approved cleanup record

Cost control and rollback

Estimate gateway hours, VM compute and disk, public IP, data transfer and log storage before deployment. Capture the estimate and approved ceiling rather than treating an unverified price as fixed. If acceptance fails, revoke test access and restore the last working configuration. At lab end, export evidence and obtain owner confirmation before removing only the dedicated prototype resources; verify retained disks, IPs and monitoring resources for ongoing charges.

Release record

Subscription and tenant: to record. Selected SKU and estimate: to approve. Deployed resource IDs and date: pending. Test executor and results: pending. No cloud deployment or billed service has been created by this document.

References and linked artefacts

[S1] HQ_Network_Addressing_Artefact.md, project deliverables, evidence dated 13 September 2026.

[M1] Microsoft Learn. Create and manage a VPN gateway. <https://learn.microsoft.com/en-us/azure/vpn-gateway/tutorial-create-gateway-portal> (accessed 23 September 2026).

[M2] Microsoft Learn. Configure P2S VPN gateway for Microsoft Entra ID authentication. <https://learn.microsoft.com/en-us/azure/vpn-gateway/point-to-site-entra-gateway> (accessed 23 September 2026).